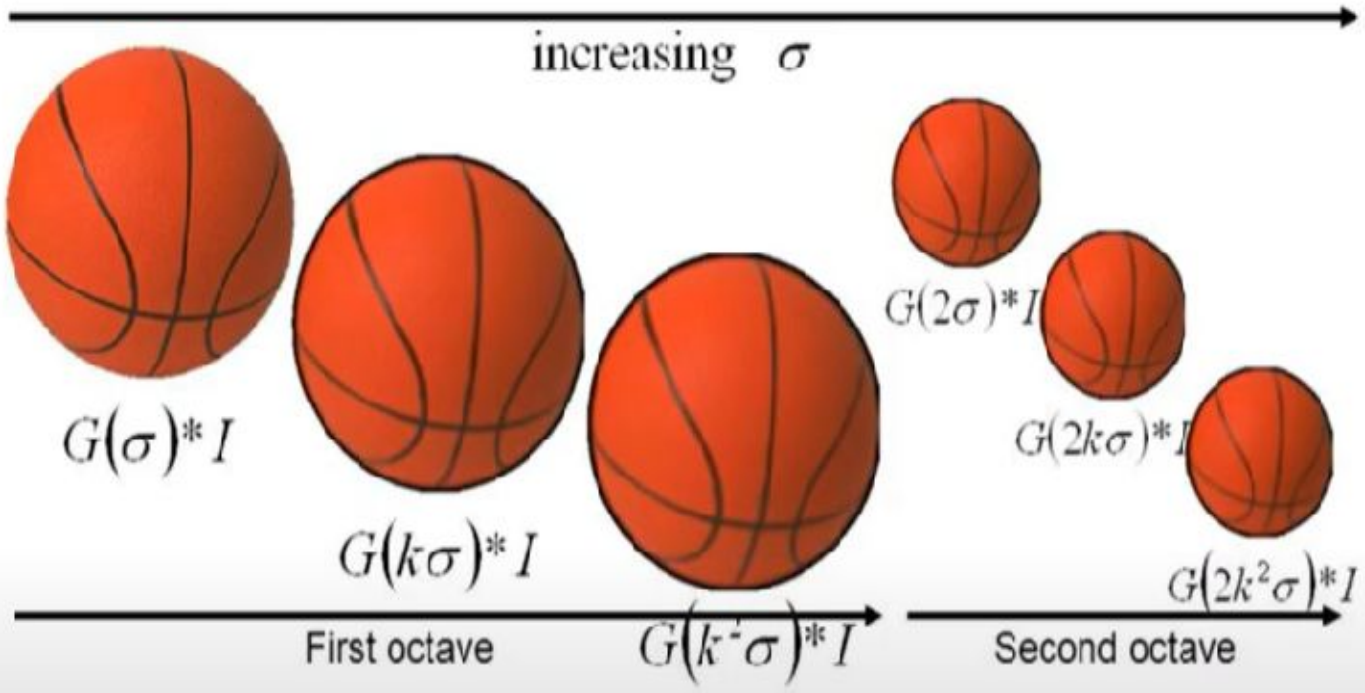
Overview of SIFT

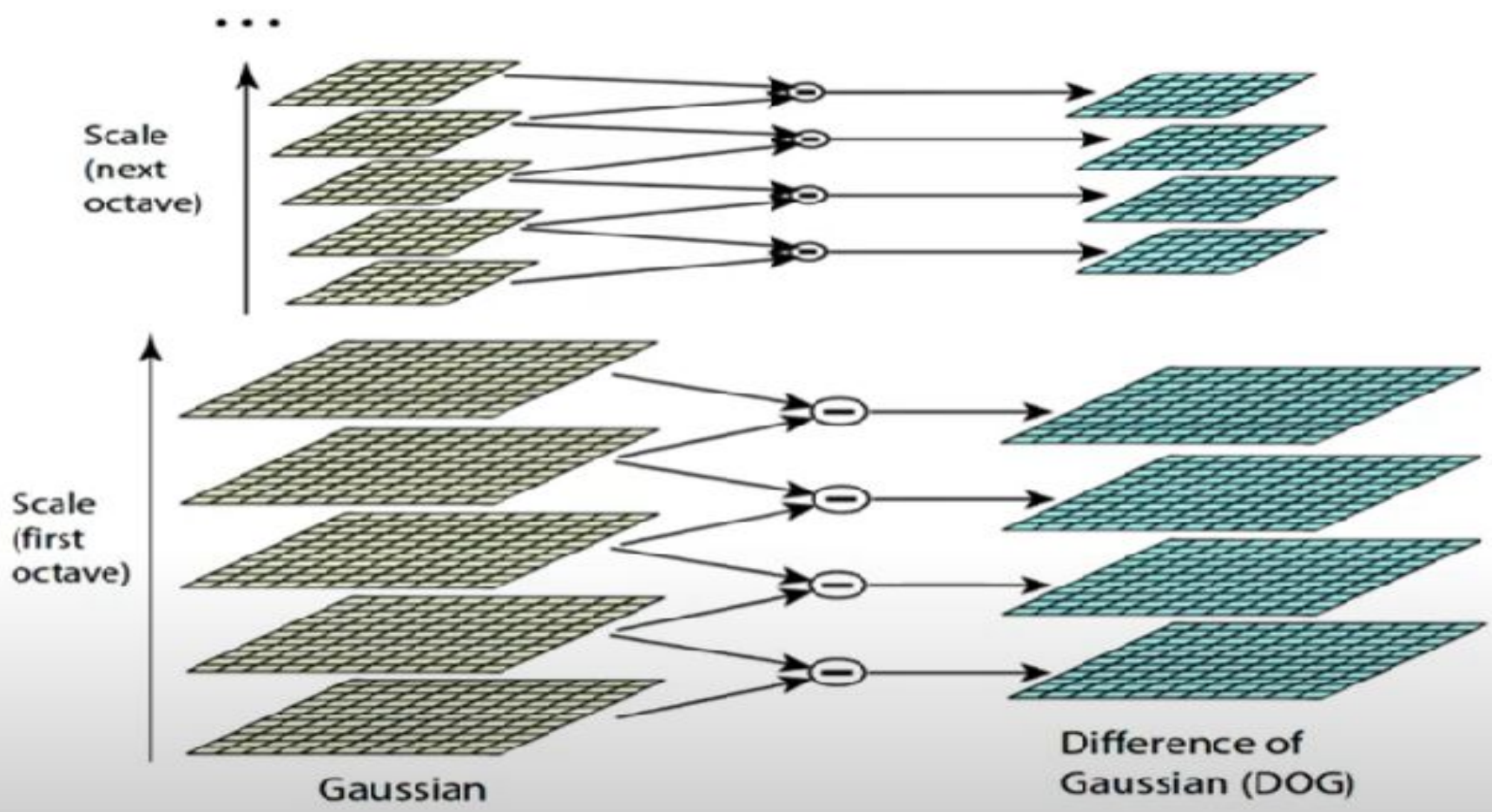
SIFT

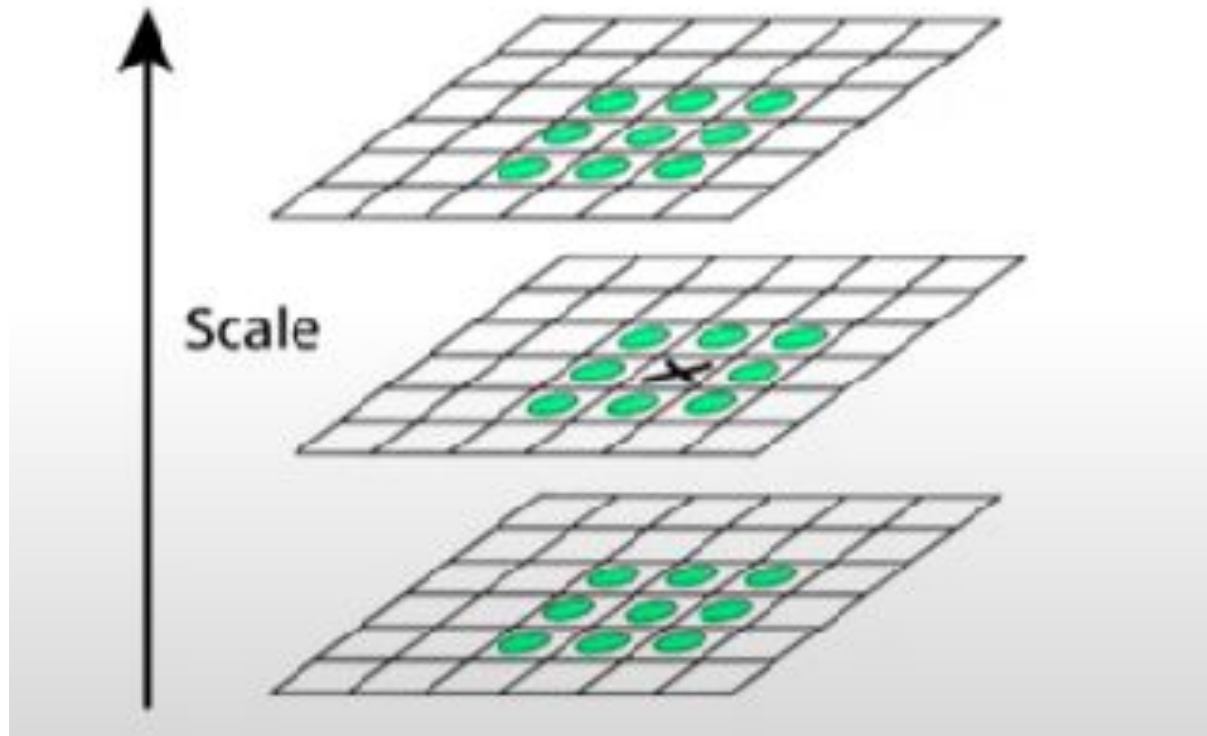
Constructing Scale Space



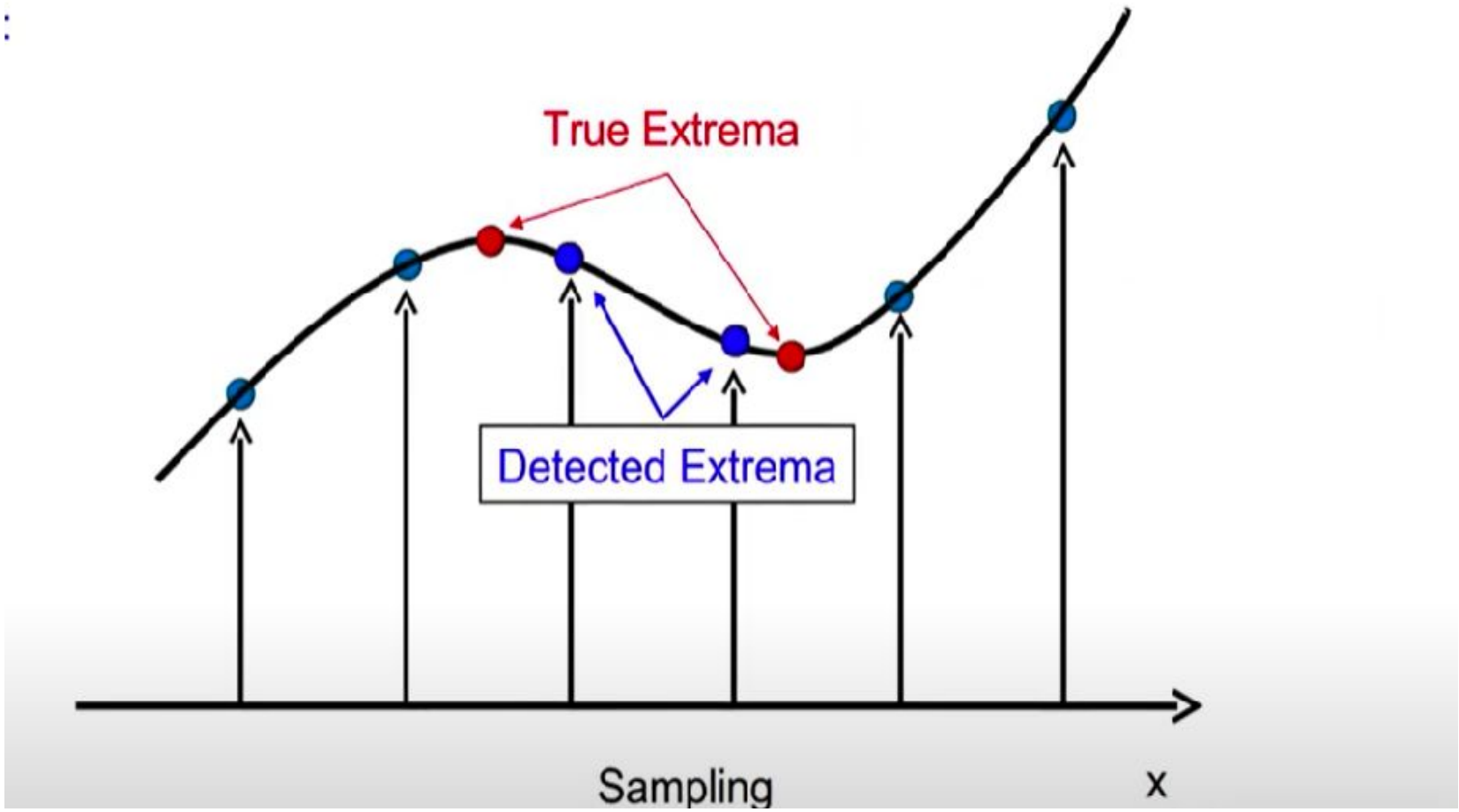
SIFT: Scale-space Extrema Detection

Difference of Gaussians



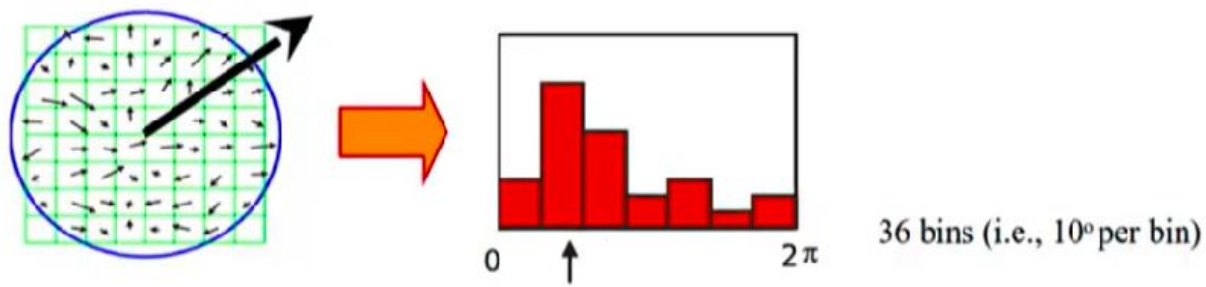

Scale

Key point localization



SIFT: Orientation Estimation

- Create histogram of gradient directions, within a region around the keypoint, at selected scale:



36 bins (i.e., 10° per bin)

SIFT: Keypoint Descriptor

- Compute gradient at each pixel in a 16 × 16 window around the detected keypoint, using the appropriate leve as detected.

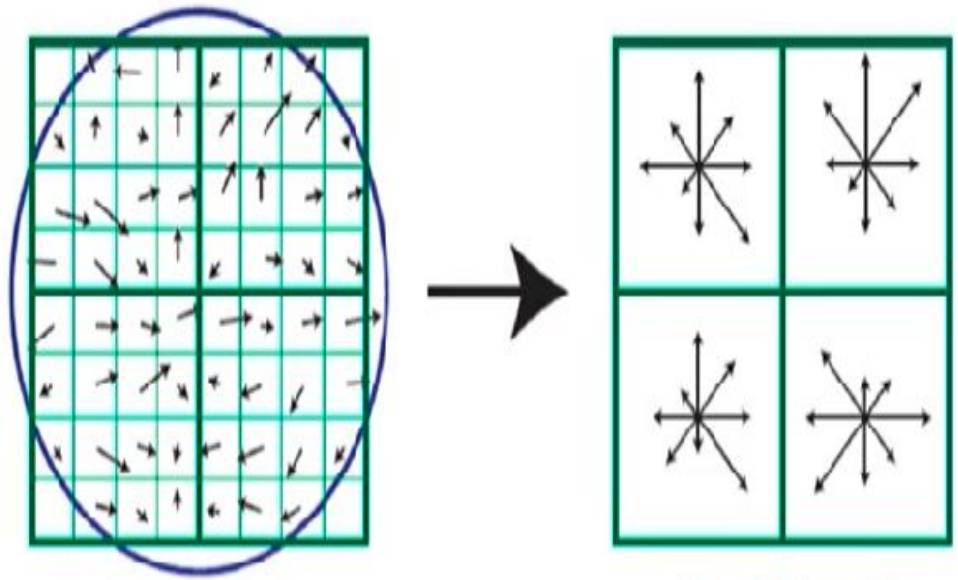
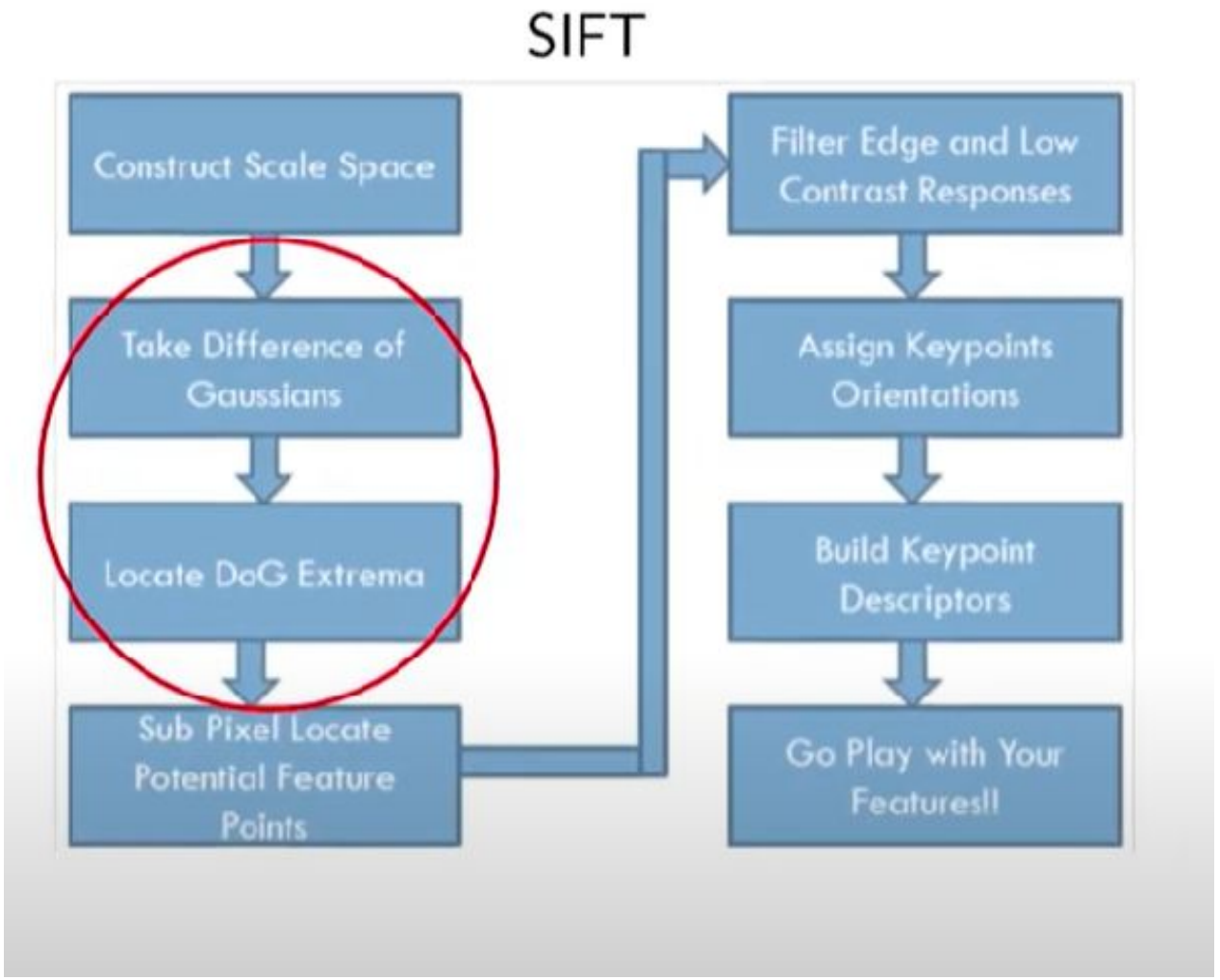


Image gradients

Keypoint descriptor

SIFT Pipeline

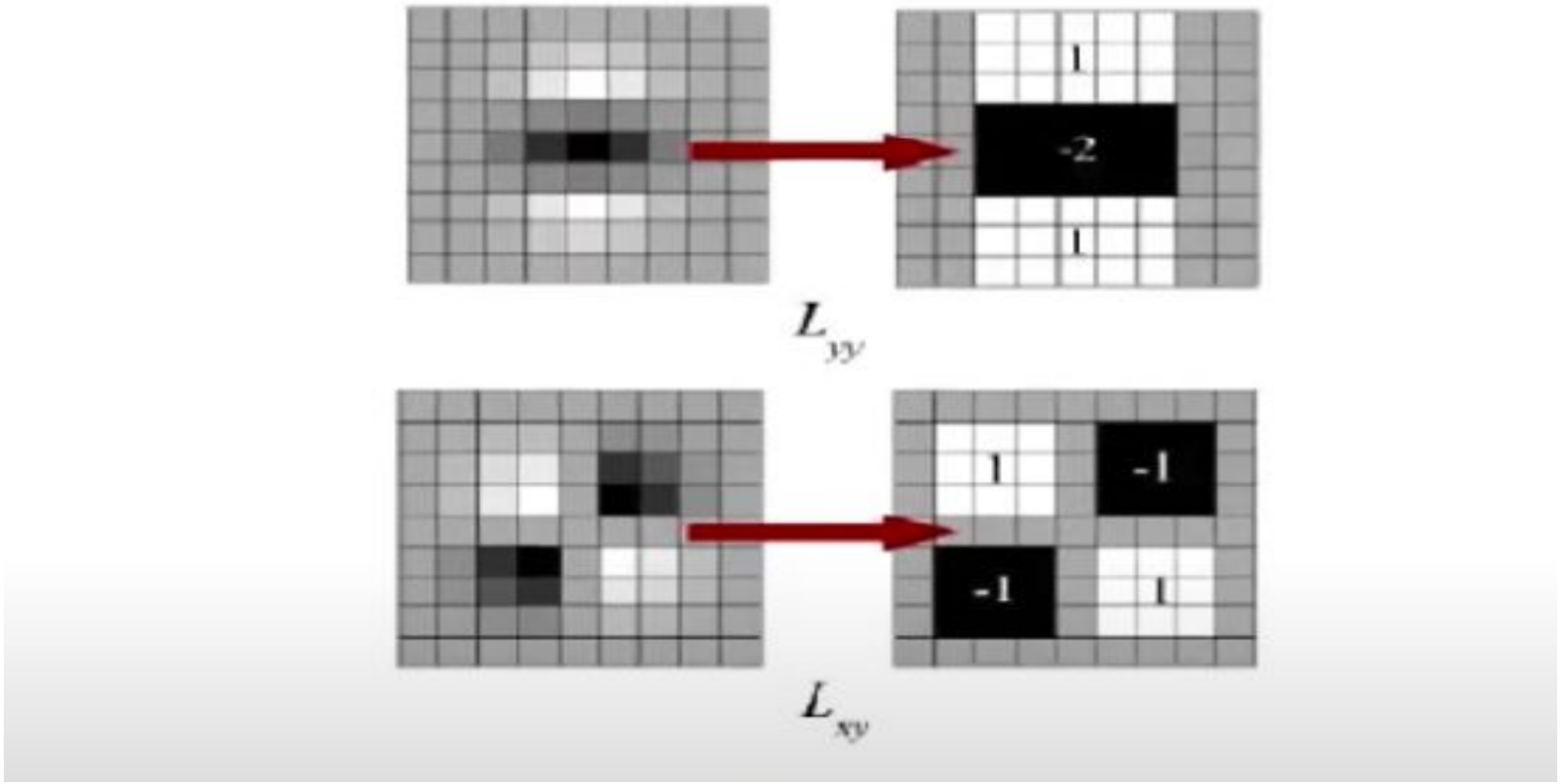


SURF

- Speeded Up Robust Features
- Uses Box filters
- Use haar wavelets to get keypoint orientations
- Simple filters – used to find gradients in x and y direction
- Good in handling blur and rotational variations
- 3 times faster than SIFT
- Considering I llumination and viewpoint changes - SURF not good
- Using Haar filters – computations can be made faster

Box filter

- Uses box filters instead of Gaussians to approximate Laplacians



GLOH

- Gradient Location orientation histogram
- Variant of SIFT
- Uses log polar binning structure
- Uses 17 spatial bin & 16 orientation bins
- Representation – 272D histogram
- Efficient – use PCA
- Reduces 272D to 128D

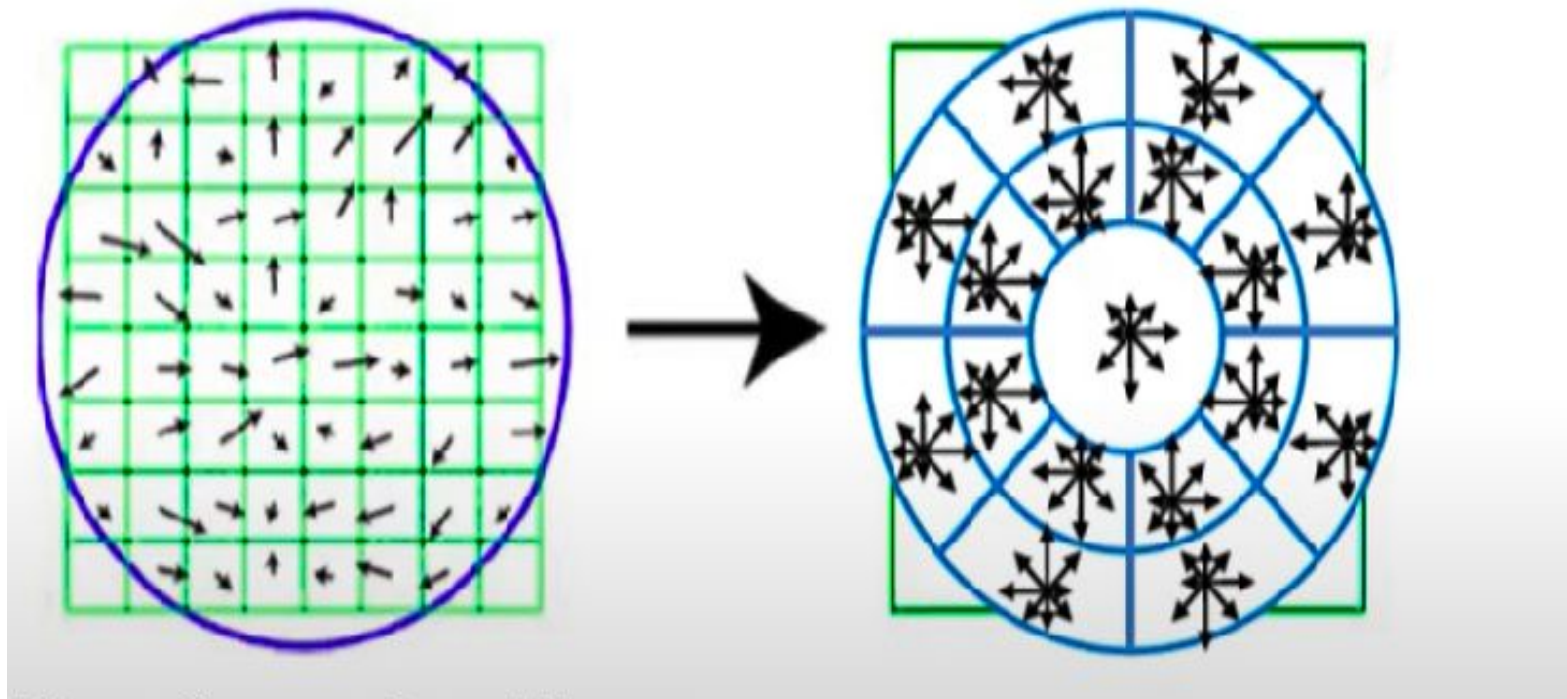
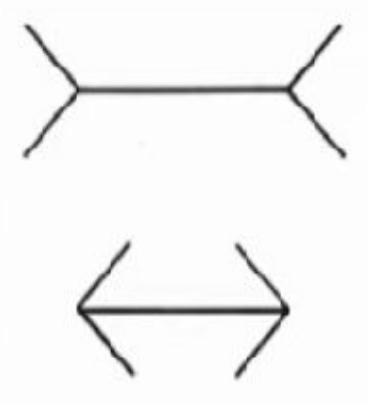


Image segmentation

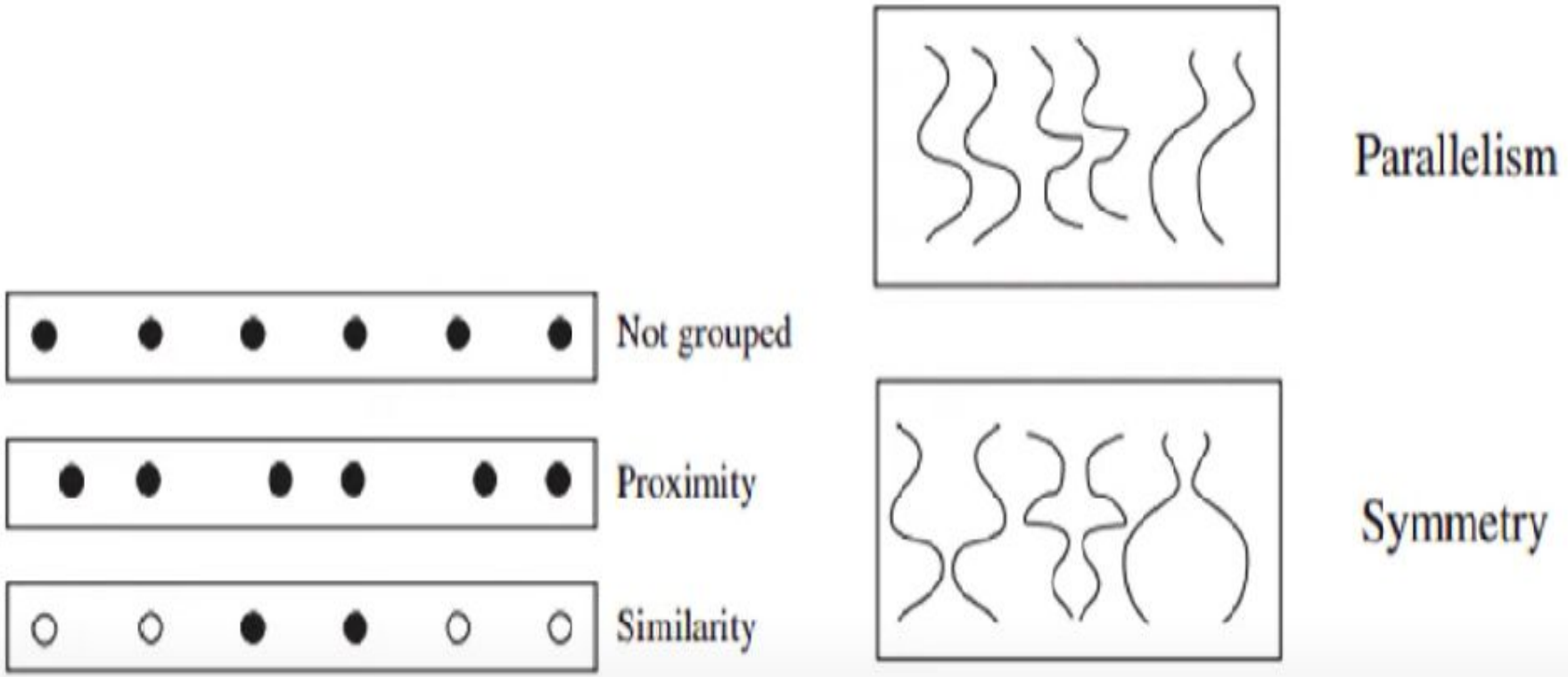
Gestalt theory

- One of the perspective of human vision
- “The whole is greater then the sum of its parts”
- Idea of grouping – understand human vision

Muller – Lyer illusion



Factors – results in grouping

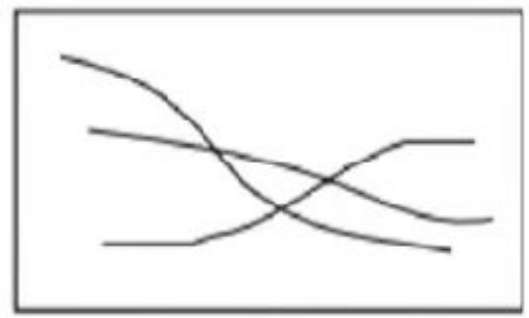




Similarity



Common Fate

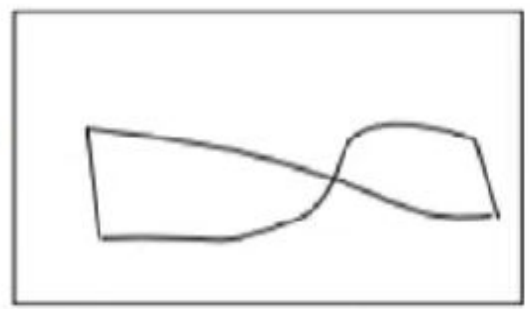


Continuity



Common Region





Closure

- Gestalt theory – fairly descriptive
- Rules are insufficiently defined
- Clustering – Group similar pixels in to different groups

Region growing

- Groups pixels or sub-regions to larger regions
- Principle of similarity
- Main segmentation criteria – homogeneity
- Important characteristics – gray level, color, texture, shape, model

Steps

- Selection of initial seed
- Seed growing criteria
- Termination of segmentation process

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

Threshold value = 4

1	0	7	8	7
0	1	8	9	8
0	0	7	9	8
0	1	8	8	9
1	2	8	8	9

Threshold value = 4

$$|f(x,y) - f(x',y')| <= T$$

B	B	A	A	A
B	B	A	A	A
B	B	A	A	A
B	B	A	A	A
B	B	A	A	A

Split and Merge algorithm

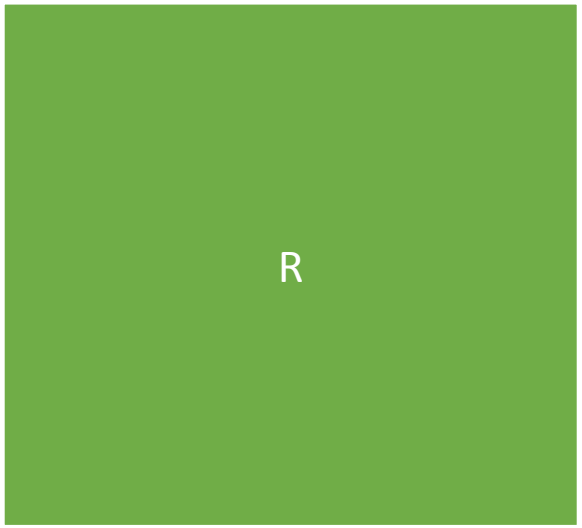
- Alternate method
- Image – subdivided in to arbitrary disjoined region
- Consider different characteristics
- Observe the discontinuity – start subdivision
- Two phases
- Split
- Merge

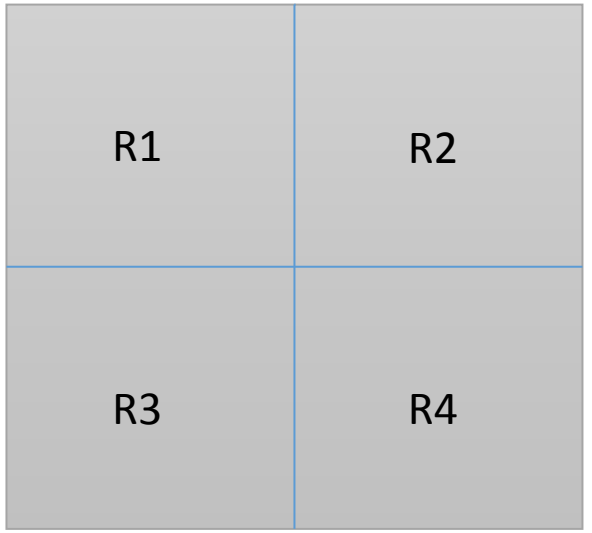
Split & Merge Algorithm

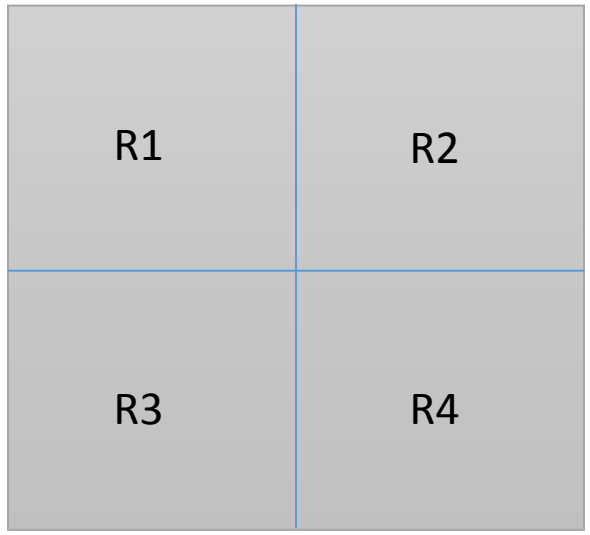
- Entire image – single region
- Observe the homogeneity
- If not met , split into 4 quadrants equally
- Repeat the process unless the condition is met
- Regions are merged – common properties are shared

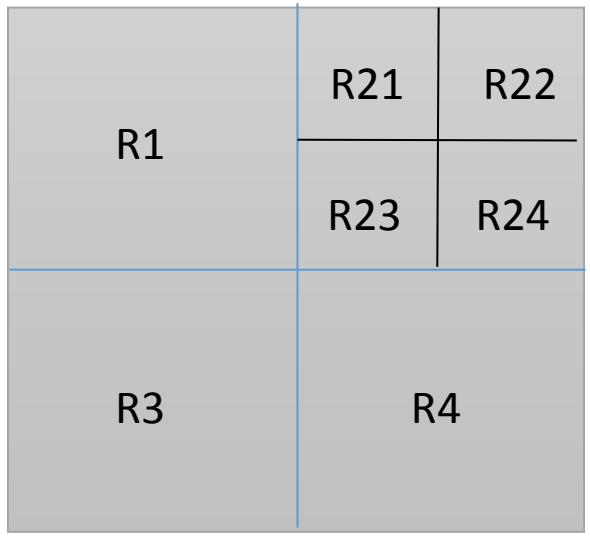
Two Phases

- Phase I : Split and continue subdivision until stopping criteria is fulfilled
- Stop when no further splitting is not possible
- Otherwise continue Phase I
- Phase II : Merge adjacent regions if the regions share any common criteria
- Stop when no further merging can be done
- Otherwise Phase II is continued
- Also known as “ Divide and Conquer” or “Top Down Approach”.


R

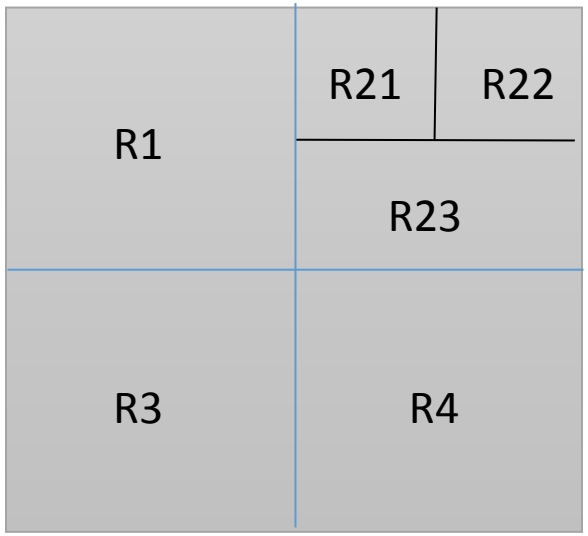

R1
R2
R3
R4


R1
R2
R3
R4


R1
R21
R22
R23
R24
R3
R4

R23 and R24 share common characteristic

- So merging can be done



Quadtree of splitting and merging

Example

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Example

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Example

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Example - Merging

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Example - Merging

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

Example - Merging

6	5	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4